

SGMS2 (Q8NHU3) — "is_active_in ER membrane" (GO:0005789): Function-Assignment Review

Hypothesis slug: wild-type-er-activity-and-export-signal **Focus type:** function_assignment

Question: Does wild-type human SGMS2 (SMS2, Q8NHU3) — and by extension SGMS1 (SMS1, Q86VZ5) — physiologically execute its catalytic (sphingomyelin synthase) function **at the ER membrane** (GO:0005789, `is_active_in`), separately from ER transit and disease-variant retention?

Executive Judgment

Verdict: Refuted for wild-type physiological ER activity (equivalently: the ER-membrane `is_active_in` annotation is over-annotated / a paralog IBD carry-over).

Target-specific human evidence does not support wild-type SMS2 catalysis at the ER. On the contrary, the most direct evidence — an **autonomous ER-export signal** in SMS2 — shows the protein is actively removed from the ER, and its physiological sphingomyelin (SM) synthesis occurs downstream at the **trans-Golgi and plasma membrane**. ER-localized catalysis is observed **only** for export-defective pathogenic variants (p.Ile62Ser, p.Met64Arg), where it is a **pathological gain of location**, not the wild-type function. The ER-positive PANTHER IBD ancestral node (PTN000480004) is anchored by the genuinely ER-resident sibling **SMSr/SAMD8** (a ceramide-phosphoethanolamine synthase) and a fly ortholog; propagating that ER activity to the SMS2/SMS1 leaves is classic paralog over-annotation.

Most important caveats. (1) This is a location-of-function judgment, not a catalytic-capacity judgment: when artificially/pathologically retained in the ER, SMS2 *can* catalyze SM synthesis there ("retain full enzymatic activity"). The GO `is_active_in ER membrane` claim for the **wild-type** gene product nonetheless fails because physiological activity is spatially restricted away from the ER by the export signal. (2) SGMS1 has its own (Golgi) localization evidence and should not be assumed identical to SGMS2; neither paralog has positive wild-type ER-activity

evidence. (3) The seed's request to treat the trafficking data as "target-specific evidence to be assessed, not dismissed" is honored — and when read fully, that trafficking evidence argues *against* wild-type ER activity.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context
P 36102623 (Sokoya, Parolek, Holthuis et al., <i>eLife</i> 2022)	Direct assay + mutant phenotype + trafficking (SMSr/SMS2 chimera, Fig 1)	Refutes WT ER activity / qualifies	Where does WT SMS2 execute catalysis; is ER activity WT or variant-specific	WT SMS2 carries an autonomous ER-export signal and leaves the ER; severe pathogenic variants (I62S, M64R) "retain full enzymatic activity but fail to leave the ER owing to a defective autonomous ER export signal," accumulating SM in the ER. Physiological SM production is in the trans-Golgi .	Human SMS2 chimeras; patient-derived fibroblasts
P 30779713 (Pekkinen et al., <i>JCI Insight</i> 2019; also fig cited by seed)	Mutant phenotype + localization	Refutes WT ER activity / qualifies	Origin of ER SM synthesis	SMS2 is described as plasma-membrane-resident ; p.Ile62Ser and p.Met64Arg "enhanced the rate of de novo sphingomyelin production by blocking export of a functional enzyme from the endoplasmic reticulum "; p.Arg50 is <i>catalytically inactive</i> *.	6 families; osteoporosis / spondylometaphyseal dysplasia
P 14685263 (Huitema, van den Dikkenberg, Brouwers,	Localization / functional cloning	Refutes ER is_active_in	WT physiological site of SMS2/ SMS1	"human SMS1 is localised to the Golgi, SMS2 resides primarily at the plasma membrane"; "SM synthesis occurs in the lumen of the	Human/mouse elegans; yeast complementation

Citation	Evidence type	Stance	Claim tested	Key finding	Context
Holthuis, <i>EMBO J</i> 2004)				Golgi as well as on the cell surface."	
P 19506037 (Vacaru et al., <i>J Cell Biol</i> 2009)	Localization / direct assay	Refutes (for SMS1) / competing	Where SMS1 makes SM; identity of ER member	Ceramide is made on the cytosolic ER surface and "transported by ceramide transfer protein to the Golgi for conversion to sphingomyelin (SM) by SM synthase SMS1"; SMSr makes CPE in the ER lumen (~300× less product).	Human/mamm cells
P 24259670 (Tafesse/ Holthuis)	Localization / mutant	Competing / provenance of ER term	Identity of ER-resident family member	"SMSr (SAMD8), an ER-resident ceramide phosphoethanolamine (CPE) synthase"; ER ceramide-homeostasis sensor.	Mammalian ce
P 21980337 (Subathra et al.)	Localization	Refutes ER is_active_in	WT localization of SMS1 & SMS2	"SMS1 localizes at the Golgi while SMS2 localizes both at the Golgi and the plasma membrane."	Mammalian ce
PANTHER PTN000480004 (ER-positive IBD ancestral node) → leaves SMS2 PTN002501709,	Evolutionary / computational (IBD)	Competing / source of over-annotation	Whether ER activity should be inherited by SMS2/SMS1	ER-positive call is anchored by SAMD8/SMSr (Q96LT4) — a genuinely ER-resident CPE synthase (P 19506037)	Ortholog infer not a target-sp assay

Citation	Evidence type	Stance	Claim tested	Key finding	Context
SMS1 PTN002501710				P 24259670 — and fly FBgn0052380 . SMS1/ SMS2 evolved Golgi/ PM residence (SMS2 via an ER-export signal), so IBD inheritance mislocates the function.	)

GO Curation Implications (leads — require curator verification)

Lead 1 — Do not assert wild-type ER-membrane `is_active_in` for SGMS2. - CC (location of function): GO:0005789 (endoplasmic reticulum membrane) with `is_active_in` should be **removed / not asserted** for wild-type SGMS2. If any ER relationship is kept, it should at most be a transient `located_in` reflecting biogenesis/transit — which is not a function-execution statement and is generally not curated as `is_active_in`. - **Preferred CC `is_active_in` for SGMS2:** GO:0005886 (plasma membrane) and GO:0000139 (Golgi membrane) / trans-Golgi network membrane.

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P 36102623

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- **Preferred CC `is_active_in` for SGMS1:** GO:0000139 (Golgi membrane). Do not copy an ER term from SGMS2 or from the IBD node. - **MF (unchanged, not the focus):** GO:0033188 sphingomyelin synthase activity. The seed explicitly asks not to re-litigate PC-vs-CDP-choline chemistry; MF is retained. - **Disease-variant ER activity** (I62S, M64R) is a **variant/pathological** observation and should not become a wild-type gene-product CC annotation. If captured, it belongs in variant/phenotype curation (GO-CAM disease model), not as `is_active_in` ER membrane for the reference gene product.

Lead 2 — Flag the IBD node (confirmed by QuickGO evidence codes, Iteration 3). Direct QuickGO retrieval shows the ER-membrane term on the human leaves is **IBA-only**, propagated from the ancestral node whose ER call is experimentally anchored by the ER-resident sibling SAMD8/SMSr and the fly ortholog. Curators should down-weight this IBA-only ER support and rely on the direct experimental (IDA) evidence, which places SGMS2/SGMS1 at Golgi/PM.

GO decision table (grounded in current QuickGO evidence codes; leads requiring curator verification):

Gene	GO term	Current evidence	Experimental support for ER?	Curation lead
SGMS2 Q8NHU3	GO:0005789 ER membrane	IBA only (GO_REF:0000033; with = FBgn0052380, PANTHER PTN000480004, Q96LT4/SAMD8)	None	Remove / do not assert is_active_in ; ER is transit-only at most
SGMS2 Q8NHU3	GO:0000139 Golgi membrane	IDA P 14685263 (+ IBA/IEA)	Direct	Retain (is_active_in / located_in)
SGMS2 Q8NHU3	GO:0005886 plasma membrane	IDA P 14685263 P 30779713 TAS Reactome	Direct	Retain (is_active_in / located_in)
SGMS1 Q86VZ5	GO:0005789 ER membrane	IBA only (same with- list); old TAS GO:0005783 whole-ER P 14976195	None (membrane-level)	Remove / do not assert is_active_in
SGMS1 Q86VZ5	GO:0000139 / GO:1990676 Golgi (trans-cisterna) membrane	IDA P 14685263 P 30242129 EXP P 17449912	Direct	Retain
SAMD8 Q96LT4 (SMSr)	GO:0005789 ER membrane	IDA P 19506037 (+ IEA, TAS)	Direct	Retain — correct here; the true ER member / IBD donor

Provenance artifacts: `/tmp/SGMS2_ER_evidence_matrix.csv`, `/tmp/SGMS2_GO_decision_table.csv` (generated from live QuickGO annotation queries this run).

Avoid "protein binding" as an endpoint here — the informative MF/CC terms above are supported.

Mechanistic Scope

- **Immediate molecular function:** transfer of phosphocholine from phosphatidylcholine onto ceramide → sphingomyelin + diacylglycerol (sphingomyelin synthase; GO:0033188). Catalytic site faces the **luminal/exoplasmic** leaflet.
- **Physiological location of that function:** trans-Golgi lumen (SMS1, SMS2) and cell-surface/plasma membrane (SMS2). SM production in the trans-Golgi builds the SM/sterol gradient of the secretory pathway

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P 36102623

intro).

- **What is downstream / not direct gene-product function at the ER:** (a) the ER SM accumulation, disrupted transbilayer SM asymmetry, and altered cholesterol/glycerophospholipid order seen in patient cells are **consequences of variant ER retention**, not WT ER catalysis; (b) osteoporosis/skeletal dysplasia are organ-level disease manifestations, not molecular-function locations.

Conflicts and Alternatives

1. **"Retained variants are active in the ER, so the enzyme *can* act there."** True but non-physiological. `is_active_in` for a reference gene product should reflect where the **wild-type** enzyme acts; the export signal restricts that to post-ER compartments. Catalytic *capacity* ≠ physiological *location of activity*. (The seed makes exactly this distinction.)
2. **Paralog confusion (the dominant alternative).** SMSr/SAMD8 is legitimately ER-active (CPE synthesis). An ER annotation is correct for SAMD8 and is the likely IBD source; it should not be transferred to SMS2/SMS1.

3. Arg50* interpretation.

P 30779713

calls p.Arg50 *catalytically inactive*;

P 36102623

Discussion qualifies this via possible alternative translation initiation (explicitly unpublished). This nuance concerns the loss-of-function variant, not the ER-activity question, and does not change the verdict.*

4. **Secretory transit ≠ catalysis.** All secretory-pathway membrane enzymes fold in and traverse the ER; transit alone is not evidence of ER `is_active_in`. No study reports meaningful WT SM synthesis in the ER.
 5. **Isoform/paralog specificity.** SGMS1 evidence is Golgi-specific; do not assume SGMS1 = SGMS2. Neither has positive WT ER-activity data.
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Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Direct measurement of WT SMS2 SM-synthase activity <i>in the ER</i> vs Golgi/PM	Abstracts of P 36102623 / 30779713/14685263	Confirms activity is spatially excluded from ER in WT	Organelle-resolved SM-synthase assays / proximity biosensors on WT vs export-defective SMS2 (partly done in 36102623 — full-text figures would quantify)
Exact provenance of the ER annotation in the GO record	Inferred as PANTHER IBD from node PTN000480004	Determines whether removal is a clean IBD down-weight or overrides experimental evidence	Curator inspection of the SGMS2 GO evidence codes and GO-CAM model
SGMS1 (Q86VZ5) trafficking / any residual ER function	Now checked: Golgi localization confirmed independently (P 14685263) (P 21980337) and SM synthesis assigned to the Golgi (P 19506037); ER member is SMSr/SAMD8	Seed warns paralogs may differ — addressed: SGMS1 also lacks WT ER-activity evidence	Direct SMS1 organelle-resolved activity assay would fully close it, but current evidence already argues against ER <code>is_active_in</code> for SGMS1
Topology detail (leaflet facing) for SMS2 at PM vs Golgi	Not independently verified here	Supports luminal/exoplasmic catalytic orientation consistent with Golgi/PM function	Cited in family literature; full-text/topology papers

I could not fetch every abstract via automated PubMed search (the endpoint returned no hits for some multi-term queries); I therefore retrieved the four decisive papers directly by PMID via NCBI E-utilities and quote them verbatim above. I did not access the eLife or JCI Insight full-text figures programmatically; figure-level claims (Fig 1 chimera design) are taken from the seed and the abstracts and are labeled as such.

Discriminating Tests

1. **Organelle-resolved catalysis assay:** compartment-specific SM biosensors (e.g., EGFP-lysenin / EqtSM) or in-vitro SM-synthase assays on purified ER vs Golgi vs PM fractions, comparing WT SMS2 to an export-signal-ablated construct. Prediction: WT shows Golgi/PM activity, negligible ER; export mutant shows ER activity.
2. **Export-signal swap (already the crux of**

P 36102623

confirm that grafting the SMS2 ER-export signal onto ER-resident SMSr relocalizes it, and that removing it retains SMS2 in the ER — directly separating "transit" from "activity."
3. **Retention-clamp control:** KDEL/ER-anchor-tagged WT SMS2 to demonstrate that ER catalysis is achievable but requires forced retention (i.e., non-physiological).
4. **SGMS1 parallel experiments** to test the paralog-difference assumption explicitly.

Curation Leads (require curator verification)

- **Candidate action change:** Change/remove the wild-type SGMS2 `is_active_in` GO:0005789 (ER membrane) annotation; treat ER involvement as transit-only (`located_in`, transient) or as variant-specific pathology, not core function.
- **Candidate replacement CC terms (SGMS2):** GO:0005886 (plasma membrane) `is_active_in`; GO:0000139 (Golgi membrane) `is_active_in`.
- **Candidate CC term (SGMS1):** GO:0000139 (Golgi membrane) `is_active_in`; do not add ER by paralog inheritance.
- **Retain (not the focus):** GO:0033188 sphingomyelin synthase activity (MF).
- **Candidate references to verify with exact snippets:**

- P 36102623**

— "SMS2 variants linked to the most severe bone phenotypes retain full enzymatic activity but fail to leave the ER owing to a defective autonomous ER export signal."
- P 30779713**

— variants "enhanced the rate of de novo sphingomyelin production by blocking export of a functional enzyme from the endoplasmic reticulum"; "p.Arg50* mutation yielded a catalytically inactive enzyme."
- P 14685263**

— "human SMS1 is localised to the Golgi, SMS2 resides primarily at the plasma membrane."
- Suggested curator questions:** (1) Did the ER-membrane term originate from PANTHER IBD node PTN000480004? (2) Is there any experimental (non-IBD) evidence code for ER `is_active_in` on SGMS2/SGMS1? If not, down-weight. (3) Should disease-variant ER activity be modeled in a separate GO-CAM disease context?
- Suggested experiments:** organelle-resolved SM-synthase assay (WT vs export-defective SMS2); SGMS1 trafficking/activity mapping.

One-line answer

For wild-type human SGMS2 (and SGMS1), physiological catalysis occurs at the Golgi/plasma membrane, not the ER; the ER-membrane `is_active_in` (GO:0005789) annotation is unsupported by target-specific evidence and is best explained as a PANTHER-IBD carry-over from the ER-resident paralog SMSr/SAMD8, with true ER catalysis seen only as a pathological consequence of export-defective disease variants.